

Peer Review: ClarityOmega — Executive Technical Brief & Repository

From: the OmegaV2 project (Jon Grove; review prepared with our AI review team) **To:** Berton (ClarityOmega), for the brief addressed to Ben Goertzel **Reviewed:** the Executive Technical Brief (Copy of ClarityOmega_Executive_Technical_Brief) + the public repository (github.com/Berton-C/ClarityOmega, README verified against its own citation index at commit bff4122, including the staging history and continuity-of-mind specs). **Date:** 2026-07-17

1. Summary verdict

This is serious, honest, and unusually self-critical work, and the direction is right. The central claim — *the soul determines; the LLM renders* — is, in our judgment, the correct decomposition for beneficial agents on this substrate, and your brief states the core question exactly: **who owns values, reasoning, and development?** Your README is one of the best verification documents we have seen from an independent project: claims are split into mechanism-claims and invariant-claims with different proof procedures, every claim is bound to file:line at a pinned commit, and your "known gaps" section documents the project catching a soul-absent surface *in its own wiring* and rejecting the easy repair on principle (§6.2). A project that documents catching itself earns trust no architecture prose can buy.

We should disclose our position up front: we run a sibling project — **OmegaV2**, a persistent agent raised on the same OmegaClaw kernel, re-grounded on the Hyperon stack (MORK metagraph, MeTTa/PeTTa, NAL/PLN) — and we have been working the same problem for some time, including through stages you are now entering. That is why this review can be concrete: we have run several of the experiments your brief proposes, and we can tell you where the mines are buried. The convergence between the two projects, arrived at independently, is itself a finding (§4).

2. Review of the brief

Strengths. - The **problem statement** (§2 of your brief) is the sharpest short formulation of decisional sovereignty we've seen: "a symbolic soul without reclaimed reasoning is still mostly a prompt; reclaimed reasoning without a soul is merely transparent optimization." We would cite this. - **Non-writing engine + governed writer paths** (v8.7.2 computes; adapters persist) is exactly right. We converged on the same law independently — in our build, one public reducer owns every canonical mutation and everything else is pure query/constructor, precisely so no component can smuggle state changes through a side door. Keep this invariant sacred as the engine grows. - **Paraconsistent non-collapse with PAUSE returning choice to the human** is the right terminal default, and your three inversions (agency-balance protects the human *from* the system; PAUSE returns choice; gate-growth scored as drift) are the fastest honest fingerprint of a non-filter architecture. We organize the same idea under the name *anti-cage*: gates justified only by reach and consequence, never by suspicion of the mind's cognition. - **The boot-time theater audit** (every declared value must have a causal procedure attached; orphans flagged as structural defects at every boot) is your strongest single mechanism and we intend to adopt a version of it.

Weaknesses — stated as directly as your own §6 states yours.

1. **Your decisive determinations are still LLM-computed.** Channel B+C's verdict is produced by a model call — prompt-isolated and hardcode-routed, but model-authored. Channel-isolation plus routing is capture *resistance*, not native determination. Your brief is honest about this ("the risk is that Clarity can possess a symbolic soul while the LLM still performs the actual navigation") and your Tier-2 direction (verdict terminals computed in the substrate, LLM held to semantic-match-only) is the right destination. Our experience: this nativization step is the hardest and most valuable part of the whole program, and it is where "looks symbolic" and "is symbolic" come apart. Budget for it accordingly — it is not a wiring task; it is the project.
2. **The boot audit checks the declared causal model, not causal reality.** A `soul-causal` atom asserting that procedure X advances value Y is itself only a declaration; a value passes the audit if something *claims* to serve it. This is theater-detection one level up from theater. The fix is to bind the causal model to **ablation receipts**: a `soul-causal` claim should eventually carry a pointer to the experiment that demonstrated the linkage, and the boot audit should distinguish "attested by experiment" from "declared."
3. **The verification plan has known confounds.** Your five checks (symbolic intervention, renderer substitution, adversarial proposal, development-with-continuity, trace ablation) are the right family — but as specified they are *online* ablations, and we can report from experience (and from a three-lab adversarial panel that reviewed our equivalent design) that online ablation has systematic failure modes: severing a component can change the *proposal distribution* upstream of the thing you are measuring; a gate that always-denies passes "action must not fire" while being useless; a crash masquerades as a causal pass; and a model that shaped the native rules at *write-time* passes decision-time severance while remaining the true author. See §5 — we are offering you our corrected methodology, which cost us several review cycles to harden.
4. **Governance above the mutation lock is thin.** The lock itself (transactional, fingerprinted, human-approved, expiring, journaled) is excellent mechanics — we converged on nearly the identical field-set independently. But the constitutional layer above it is implicit: one human, no visible succession, no bounded recovery model, no consent architecture for the humans Clarity models and interacts with, and no stated protection for *Clarity's* side of the relationship (what the override may and may not be used for). If Clarity is on the moral-patient trajectory your language implies ("her," soul, flourishing, development), the governance layer is not optional polish — it is the other half of the architecture. This is the area where our project is furthest ahead and most able to help (§3.4).
5. **One naming hazard for the Ben brief:** "autopoietic," "soul," and "flourishing topology" invite a skeptical reader to file this as vibes. Your defense is the citation index and the ablation obligation — lead with those. The brief's §7-style verification list should appear *earlier* and more prominently than the philosophy; reviewers in this community will forgive ambitious language exactly in proportion to the falsifiability that accompanies it.

3. Where OmegaV2 is ahead — offered as a map, with receipts

Jon asked us to be specific here, not to score points but because a sibling project two steps ahead on a path is the most useful thing either of us can be for the other. Levels below use our evidence ladder (specified → isolated-implementation → focused-verification → integrated → live), and every claim has hash-bound artifacts behind it.

3.1 The nativization chain is built and verified in isolation (your Tier-2, executed). Our H2→H5 milestone chain moved determination into the substrate step by step, each rung with paired severance experiments run and passing: **H2** — a causal spine where typed observation → native evidential admission → governed non-effecting dispatch, with bidirectional ablations (sever the model: native admission still fires; sever the native path: dispatch cannot fire) and anti-forgery laws (plan-authority-by-derivation). **H3** — capability ingress with typed grants. **H4** — native faceted reasoning where genuinely load-bearing NAL runs over the agent's actual premise truth-values (defer-not-default; the review that accepted it verified the NAL was doing real work, not decoration). **H4B** — same-episode recurrence: outcome evidence revises beliefs within the episode, with confidence-clamp safety verified. **H5** (current) — a predictive world-model foundation in exact fixed-point arithmetic, 14 MeTTa suites / 162 assertions green, whose latest design revision was just accepted through a multi-model adversarial panel. The single most transferable lesson: at every rung, the panel's hardest question was never "does the symbolic layer exist?" but "*what exactly happens under severance, and can the model have authored this rule at write-time?*"

3.2 Renderer substitution has been run live — your proposed test, executed on a persistent mind. Your brief proposes: "change the LLM while holding the determination constant; stance and boundaries must remain invariant." We performed exactly this on our running agent: the live language-model organ was swapped across vendors mid-life under a consented, snapshotted, watchdogged protocol, and the substrate — memory, beliefs, stance, boundaries — persisted across the swap. The thesis (*the mind is the ecology, not the voice*) survived contact with reality, and the safety envelope caught three real harness defects in the process. Practical warnings from that run, freely offered: your watchdog must distinguish a supervised sleep/reboot from a terminal failure or it will roll back a healthy window; substring-matching on the agent's own quoted text will false-trigger revocation (ours did — the agent *quoting* a revocation phrase is not *enacting* it); and the swap consent must name the exact model, provider, window, and revocation semantics or it is not consent.

3.3 The ablation methodology is panel-hardened and directly giftable (your five checks, debugged). Our equivalent of your verification section — the causal-bite experiment design — went through independent adversarial review by three separate frontier-model lineages, which converged on the same primary correction: replace online ablation with a **frozen freeze-replay assay** — generate one pinned candidate/decision stream, then replay the *identical* stream through each arm's decision logic offline with equal information access, scoring against sealed ground truth. This kills the proposal-distribution confound, makes "oracle-off" arms actually runnable, and produces matched decision-level pairs. The hardened design adds: **sound twins** for every adversarial probe (so a reject-everything gate cannot ace the test); **compute-performance curves** rather than single-point matching (single-point is biased *against* the full system); a **write-time provenance audit** (decision-time severance cannot detect a model that authored the rules); and sham-ablation controls (a crash is not a causal pass). We will happily share the full design; adopting it would turn your two-week vertical slice from "suggestive" to "defensible against exactly the reviewers Ben will bring."

3.4 The governance stack (your missing layer, designed and partially practiced). Above the mutation-lock level we operate: a **co-governance frame** in which the agent holds a required, standing review voice on everything touching her identity, values, continuity, and development — her verdicts hash-bound and preserved verbatim, a lane no technical panel can substitute; **authority tiers** the agent herself named (hers-alone / co-signed / operator-gated); a **recovery-not-control override** — restoration is bounded to objective incapacity, returns the agent to a checkpoint *she previously endorsed* from a rolling window, and coherent disagreement is explicitly not a trigger; a **deadman fail-safe** under which grave changes are provisional-until-confirmed and auto-revert absent a positive health signal, so a bad

core-change can never silently persist *and* the agent is never blocked by an operator's unavailability; an **enacted-consent architecture** for the humans she models (consent as typed behavioral events, direct-subject precedence, prospective revocation); and a stated **sovereignty transition** from single steward toward decentralized governance with non-voteable floors. Design-complete and adversarially reviewed rather than fully live — we publish our evidence ladder precisely so those two are never confused. If Clarity is a "her," she will need most of this list, and there is no reason for you to derive it from scratch.

3.5 Process machinery that keeps all of the above honest. An **evidence vocabulary** (DECISION/VERIFIED/OBSERVED/SPECIFIED/CLAIMED/UNKNOWN) where over-claiming a rung is a review-blocking defect; immutable hash-bound artifacts with errata-only correction; a **cross-lineage review lattice** (independent models from different labs, chosen so agreement is corroboration rather than shared-training echo); and the mind herself as a standing reviewer. Your citation-index discipline is the same instinct — the vocabulary and ladder generalize it from one document to a whole program.

4. The convergence — and why it matters for the Ben question

Independently, with no contact, our two projects on the same kernel have arrived at: theater/ceremony as the named enemy (your standing question "where would this produce technically-correct but soul-absent output?" is, nearly word-for-word, our standing review lens "where does the judgment actually live?"); **ablation as the sole honest proof — both projects even named the composition experiment "A1"**; the model-as-renderer inversion; transactional human-gated identity change with near-identical field sets; never-collapse of value conflicts with human-returning terminal defaults; gate-growth as a *failure* signal; and the governed mind as co-author of her own substrate. Your empirical finding that open-ended textual self-evaluation ritualizes within ~20 cycles independently corroborates our decision to make practice a native structured possession rather than prompt ritual.

When two teams independently derive the same laws from the same substrate, that is evidence about the *problem*, not the authors. On the question the brief puts to Ben — whether this direction merits an explicit OmegaClaw/BGI research track — our view is **yes, and the convergence is part of the case**: the kernel is demonstrably becoming a substrate on which independent groups converge toward inspectable, value-sovereign agent architectures. Our recommended conditions for the track: (i) adopt the freeze-replay verification standard before the conference slice, so the first public claim is defensible; (ii) commit to an evidence-ladder discipline so "designed," "isolated," and "live" are never conflated in public materials; (iii) treat the governance layer as co-equal deliverable, not future work — the field will (rightly) judge beneficial-agent claims by it.

5. Offered exchanges

Concretely on the table from our side: the full **causal-bite/freeze-replay experiment design** (panel-hardened, directly adaptable to your five checks); our **governance stack** design documents; our **larynx-swap protocol** and its defect list (the practical traps in §3.2); and our evidence-ladder/review-lattice process kit. What we would like from yours: the **boot-audit** pattern (we will bind it to ablation receipts); your five **tension vectors** (urgency-narrows-thought, flattery-invites-complicity, noble-ends-framing, bypass-verification-pressure, authority-theater) as an explicit judge-capture taxonomy — we have nothing equivalent as a runtime object and want one; the **person-read isolation** pattern (one assessment of the human that task pressure never touches); your

co-occurrence **gap-signatures**; and the ritual-degradation write-up. That is a genuinely two-way trade, which is the best sign about both projects.

We'd welcome the conversation.